

Gauge Theory in $\mathcal{UD}$

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Abstract

This notebook extends the Relativity Toolkit (v1.9.0) from general relativity into basic gauge theory. We demystify “gauge field” by relating it to the familiar concept of reference frame. We find that forces arise from choice of gauge, specifically from the implied connections. Inertial forces, like centrifugal Coriolis force, arise from choosing a rotating reference frame. Fundamental forces, like electromagnetic and the weak nuclear forces, arise from choices of internal gauge “reference frames.” The difference is that no choice of gauge can make fundamental forces go away. Gravitational tidal forces arising from geodesic deviation are also fundamental in this sense: no choice of reference form can make them go away. In all cases, invariance of the physics, aka gauge symmetries, imply conservation laws.

In the $\mathcal{UD}$ (up-down) tensorial calculus, we implement a new, generalized covariant-derivative operator that handles both the spacetime geometry of general relativity via the Christoffel or Levi-Civita connection and internal phase geometry of electrodynamic and weak gauge connections simultaneously. We validate this engine through four automated, symbolic proofs:

- *Metric Compatibility*: the Levi-Civita connection is uniquely preserves the metric ($\nabla g = 0$)
- *$U(1)$ Gauge Covariance*: the covariant derivative of a scalar field transforms via local phase rotation
- *Field Strength Invariance*: the electromagnetic tensor $F_{\mu\nu}$ is invariant under local gauge transformations
- *Charged Vector Boson Covariance*: In the $U(1)$ sector of Weinberg-Salam electroweak theory, we show prove covariance for a vector field W^α that enjoys both spacetime and internal phase invariance

Finally, we apply the updated engine to re-derive the Riemann curvature tensor from the commutator of covariant derivatives.

- A .wl script version of code for this notebook is available at https://github.com/rebcabin/RelativityToolkit/blob/main/non_quantum_gauge_theory.wl.

Initialize the Relativity Toolkit

Optional: Quit for a Fresh Kernel

- *Note, if you call `Quit` in a notebook, evaluation may hang. If you want to “Evaluate Notebook,” mark the following two cells “Evaluatable” in the Cell→Property menu, call `Quit` and `FrontEndTokenExecute` once each, then comment out the cells, or mark them unevaluatable again. Alternatively, you can quit the kernel via the Evaluation menu (last item). At that point, you can “Evaluate Notebook” in a clean environment, or evaluate cells individually.*

```
Quit[];
```

```
In[*]:= FrontEndTokenExecute["DeleteGeneratedCells"]
```

Download the Code

```
In[1]:= RTbranch = "main";
RTbase = "https://raw.githubusercontent.com/rebcabin/RelativityToolkit/" <>
RTbranch <> "/";
RTbust = "?t=" <> ToString[Round[AbsoluteTime[]]]; (*Force fresh download*)
RTrules = RTbase <> "RelativityToolkit.wl" <> RTbust;
RTtests = RTbase <> "RelativityToolkit_RegressionTests.wl" <> RTbust;
RTviz = RTbase <> "RelativityToolkit_VisualShowcase.wl" <> RTbust;
Get[RTrules];
```

» Relativity Toolkit version : 1.9.0

» Relativity Connection Symbol : Γ

Optional: See all Tests and Results from Prior Works

Make any of the following cells evaluatable if you wish. They’re unevaluatable by default to keep this notebook trim.

```
In[*]:= Get[RTtests];
In[*]:= Get[RTviz] (* no semicolon, on purpose *)
In[*]:= Get[RTbase <> "covariant_derivative.wl"]
In[*]:= Get[RTbase <> "law_of_gamma.wl"]
In[*]:= Get[RTbase <> "connections_to_christoffels.wl"]
Get[RTbase <> "schwarzschild_from_scratch.wl"]
```

All Physics is Gauge Theory

Look at the dictionary definition of “gauge:”

1. n. A device or instrument for measuring the magnitude, amount, or contents of something. Ex.: a fuel gauge
2. n. The thickness, size, or capacity of something, especially as a standard. Ex.: the gauge of a rail line.
3. v. To estimate or determine the magnitude, amount, or volume of something. Ex. “Gauge the intrinsic luminosity of a star.”
4. v. To measure the dimensions of an object with a gauge instrument. Ex. “Gauge the assemblies and plane them to width.”

All this is refreshingly concrete, referring to ordinary objects and activities in the trades or in the physics lab. None of this hints at the abstractions of gauge theory for electrodynamics, the nuclear forces, and gravitation. Where are covariance, invariance, symmetry groups, conservation, unitarity, fibre bundles, and so on, in these definitions?

The answer is the other way around: all these abstractions are just mathematical machinery for various kinds of gauging, in the sense of measuring, various quantities in electrodynamics, the nuclear forces, and gravitation. Physics is, fundamentally, a science of measurement, and we can’t measure anything without setting up gauges. Most familiarly, we choose coordinates for measuring positions and momenta in spacetime. *Setting a gauge* is setting a coordinate basis for these quantities. Likewise, we must choose a reference for measuring the phase of an electromagnetic field. *Setting a gauge* for electrodynamics is setting that phase reference at each point of spacetime. By analogy to position and momentum, we can say that setting the phase reference is *setting a coordinate system for phase*.

It seems that coordinates, reference frames, zero points, and bases are all so deeply related, regarding purpose and necessity, that we can call all of them “gauges” and we can call the activities of setting them up as “setting gauge” or “choosing gauge.”

Now, to the surprising statement: **All Forces Arise from Choice of Gauge.**

We’re not accustomed to this idea. We imagine forces as intrinsic physics. Independently of gauges set to measure them, we get the same physics. Planets are attracted to the Sun by a gravitational force, and we get compatible results in all reference frames. Electrons float around atomic nuclei by the same attractive Coulomb force, no matter how we measure it. Springs and pulleys apply forces to masses, no matter how we measure them.

But forces change motions, and motions must be measured in coordinate systems, i.e., gauges. Different choices of gauge yield different values for motion measurements, so different gauges imply different forces to produce the observed motions. Some of these forces are only apparent, centrifugal and Coriolis forces being the canonical examples. We can make these go away by choosing different a different gauge, say an inertial frame. Other forces are such that no choice of gauge can make them go away. The electric force is like that, and so is geodesic deviation, which manifests as an unavoidable force in curved spacetime. Choice of gauge will lead to different measurements, but invariances due to unavoidable symmetries in gauge choices do reveal conserved quantities, which are the same in all gauges.

Choice of gauge can change other quantities, like phase, quantities other than force. By finding the symmetries—the choices of gauge that don’t change the physics—we find conservation laws via

Noether's theorem.

- When choice of the zero of stopwatches doesn't affect the physics, when the physics is invariant with gauge choice for time, when time symmetry obtains, then energy is conserved.
- When choice of the zero of position measurements doesn't affect the physics, when the physics is invariant with gauge choice for position, when position symmetry obtains, momentum is conserved.
- When choice of a zero for phase doesn't affect the physics, when the physics is invariant with gauge choice for phase, when phase symmetry obtains, electric charge is conserved.

And so on. Invariance to unitary transformations in nuclear physics: hypercharge and color charge are conserved.

We Hear Too Late about Gauge

In elementary physics, we don't call choice of reference frame a "gauge choice." Nor do we hear the exotic idea that gauge choice generates force. But such is the case. For an elementary example, consider measuring the altitude of an Earth satellite flying in a stable, circular, free-fall orbit. If we measure altitude with respect to mean sea level, the satellite seems to have a constant altitude, Earth's oblateness aside, and we measure no forces on it. However, if we measure altitude with respect to the irregular surface of the Earth, the satellite's altitude varies madly up and down. To naive observers, unaware of this poor choice of gauge, the satellite appears to suffer inexplicable forces pushing it up and down. However, these apparent forces arise merely from the choice of gauge. Here, the term "gauge" fits entirely with its ordinary, dictionary meaning, as an altimeter with its arbitrary, and possibly locally variable, origin setting. An even more elementary example is a rotating reference frame. If we measure positions and momenta of free particles in an inertial lab frame, we see no forces. If, however, we measure from a rotating frame, we see centrifugal and Coriolis forces, arising merely from the choice of gauge, of reference frame.

Gauge theory is totally pertinent to classical mechanics, but physics students don't study "gauge" until classical electrodynamics, where it's an abstract, *arbitrary* field whose gradient is added to the vector potential. There is no conceivable relationship of this idea to the ordinary dictionary meaning of "gauge." By that time, it's too late for our students to go back and review the forces of elementary mechanics as arising from gauge choice. Our poor students will carry multiple bags, all the way through general relativity (GR), with "gauge" a fearsome concept, until appears again as even more abstract fields in particle physics. Our students could have packed all their ideas into one piece of baggage with a concrete view of gauge as just a choice of reference frame, a frame that varies from the concrete to the abstract, but just a reference frame nonetheless.

And then, knowing that **all forces, apparent and real, do, in fact, arise from choice of gauge**, and that the symmetries of gauge choice drive conservation laws via Noether's theorem, we can put "Noether" as the travel sticker on the baggage and as the visa stamps in our passports as we journey through all of physics.

Gauge Forces as Connections

Now, with some precision, let's see how connections are the ultimate source of force from gauge choice. Recall from earlier notebooks that in GR, the Christoffel symbols are the connections that separate intrinsic curvature of spacetime, which no choice of gauge can eliminate, from extrinsic curvature caused by choice of gauge. At a single point, we can always choose a local inertial frame, a gauge that makes Christoffel symbols vanish, and acceleration, thus force, go away. This is Einstein's famous "elevator" demonstration of the Principle of Equivalence. But the paths of two particles, initially parallel, will converge or diverge in a curved spacetime, and no choice of gauge can make such "tidal forces" go away.

We've studied covariant derivative, connections, and Christoffel symbols in this series of notebooks, specifically in relativity. We've repeatedly read that we can and should recast these ideas in terms of gauge theory. Now is the time, with the clarity that gauge choice is just choice of reference frame, though in a generalized sense. Geometrical frame invariance, now geometrical gauge invariance, forces the "discovery" of covariant derivative and connection. Connection drives equations of motion, thus forces, for Schwarzschild orbits. We're now equipped to study the other forces of Nature on the same terms.

Back to our physics students, with lighter, Noether-tagged baggage, encountering gauge again, *not* for the first time, in classical electrodynamics. One realizes that the gradient of any differentiable scalar field Λ , rightfully called a gauge, can be added to the vector potential without altering the physics:

$$\bar{A}_\mu(x^\nu) = A_\mu(x^\nu) + \partial_\mu \Lambda(x^\nu) \quad (1)$$

In what sense is Λ the choice of a reference frame? Electrodynamical gauge invariance is *also* presented as **phase invariance**, a factor of $e^{i\Lambda}$ that can be multiplied into a charged matter field without altering the physics. Spoiler: choose phase in an abstract reference frame. This is the *primary* principle of electrodynamical gauge as a coordinate frame. Gradient added to vector potential is the *mechanism*, the "induced transformation," that upholds that principle. Here's how that works:

Change a matter field ψ by rotating its phase (gauge) by an arbitrary (but smoothly varying) amount, $\Lambda(x^\nu)$, at every point x^ν , without changing the geometrical coordinates x^μ :

$$\bar{\psi}(x^\nu) = e^{i\Lambda(x^\nu)} \psi(x^\nu) \quad (2)$$

This is just like choosing a coordinate system, with an arbitrary origin, locally, at each point, for the phase of the matter fields, without changing the spacetime geometry.

Take the ordinary partial derivative, ∂_μ , of this new field $\bar{\psi}$. The ordinary product rule of differentiation spits out a "garbage term" because Λ varies with position:

$$\begin{aligned} \partial_\mu \bar{\psi}(x^\nu) &= e^{i\Lambda(x^\nu)} (\partial_\mu \psi(x^\nu) + i(\partial_\mu \Lambda(x^\nu)) \psi(x^\nu)), \text{ or, abbreviating away the spacetime coordinates} \\ \partial_\mu \bar{\psi} &= e^{i\Lambda} (\partial_\mu \psi + i(\partial_\mu \Lambda) \psi) \end{aligned} \quad (3)$$

Sound familiar? This is exactly what happened in relativity. The "garbage terms" (Hessians) drove our "discovery" of covariant derivative and connection coefficients. Here, likewise, the extra term $i(\partial_\mu \Lambda) \psi$

ruins the rest of the physics.

As before, we find that the ordinary derivative is not covariant: it doesn't survive changes in phase coordinates, arbitrary changes in choice of $\Lambda(x^\mu)$.

In GR, the gauge was spacetime geometry itself, so compensating for coordinate change required 4D index manipulation. With electrodynamics, compensating for the phase changes is easier. The electrody namical connection is a one-index object because the electrody namical gauge field Λ is one-dimensional.

To clean up the electrody namical gauge garbage, we need a correction term in our covariant derivative of the matter field, just as we did with geometry and GR:

$$D_\mu \equiv \partial_\mu - iA_\mu \quad (4)$$

Yes, the connection is the vector potential, hence it drives electrody namical forces, just as the Christoffel (Levi-Civita) connection drives gravitational forces.

For the garbage terms to cancel out exactly, A_μ must transform as in Equation 1 (abbreviated):

$$\bar{A}_\mu = A_\mu + \partial_\mu \Lambda \quad (5)$$

We have the new covariant derivative

$$D_\mu \psi \equiv \partial_\mu \psi - iA_\mu \psi \quad (6)$$

and

$$\begin{aligned} D_\mu \bar{\psi} &= \partial_\mu \bar{\psi} - i\bar{A}_\mu \bar{\psi} = e^{i\Lambda} (\partial_\mu \psi + i(\partial_\mu \Lambda) \psi) - i\bar{A}_\mu \bar{\psi} \\ &= e^{i\Lambda} (\partial_\mu \psi + i(\partial_\mu \Lambda) \psi) - i(A_\mu + \partial_\mu \Lambda) (e^{i\Lambda} \psi) \\ &= e^{i\Lambda} \partial_\mu \psi + e^{i\Lambda} i(\partial_\mu \Lambda) \psi - e^{i\Lambda} iA_\mu \psi - e^{i\Lambda} i(\partial_\mu \Lambda) \psi \\ &= e^{i\Lambda} \partial_\mu \psi - e^{i\Lambda} iA_\mu \psi \\ &= e^{i\Lambda} (\partial_\mu \psi - iA_\mu \psi) \end{aligned} \quad (7)$$

transforming covariantly in gauge space with respect to the gauge choice $e^{i\Lambda}$. It so happens that A_μ is a covariant covector, so it also transforms covariantly with change in coordinates x^ν . Two covariant derivatives, one against phase, the other against geometrical coordinates. One idea of covariance.

- *We now see why it's a good idea, after all, to call the covariant derivative covariant. Such honors the metaphor everywhere without being exactly correct in GR, where the covariant derivative also pertains to contravariant indices, a fact we dismissed as "historical" in an earlier notebook.*

We'll write some new functions for the Relativity Toolkit to automate the calculation above, after just one last philosophical section, which bridges our old material for GR more deeply to the new material on gauges in general.

Covariant Derivatives, Revisited

In GR, we learned to take covariant derivatives of tensors that inhabit the tangent and cotangent tensor spaces of curved manifolds. The canonical example is the covariant derivative of a contravariant vector with respect to a basis of coordinate functions:

```
In[8]:= With[{example = HoldForm[CD[A[u[μ]], x[u[v]]]}],
  Echo[TensorForm@@example, example ""]];
```

» $(A^\mu_{;\nu}) A^\mu_{,\nu} + (A^\lambda_{;\nu}) \Gamma^\mu_{\nu\lambda}$

The connection coefficients Γ account for the twisting and turning of basis vectors in the tangent spaces as the vector A^μ is parallel-transported. In GR, this twisting and turning causes geodesics to converge or diverge, thus tidal forces. It also drives single geodesics, giving rise to ineluctable orbits, gravitational lensing, gravitational waves, and so on.

- *Emphasis: Geodesic deviation is indistinguishable from tidal force. It's not an apparent force: we can't get rid of it by a gauge choice.*

The derivative above transforms like a tensor, ensuring physics is **locally Lorentz invariant**. This invariance invites Noether's theorem, implying a conserved quantity. In GR, this conserved quantity is **spin current**, from symmetries of non-coordinate bases (see Appendix 1).

“Wallpaper” Spaces Attached to Geometry

In this notebook, we consider new spaces “attached” to points of the manifold, like wallpaper—gauge fields representing electromagnetism and nuclear forces. These wallpaper spaces have their own differential geometries above and beyond the geometry of spacetime itself.

In gravitation, gauge forces arise FROM the geometry. In other theories, gauge forces arise ON the geometry.

- *That's one reason gravitation resists unification with the other forces of Nature, as explained more fully in the appendices.*

While it is fine to analogize gauge fields as “wallpaper,” the rigorous mathematical machinery for this is that of **fibre bundles** (Appendix 2).

The objective of this notebook is to apply $\mathcal{UD}$ to these wallpaper gauge fields, encoding transformation laws, invariants, and conserved quantities with the same rigor we applied to GR.

A New CD

This new overload of CD can compute Abelian gauge connections like scalar phase. These connections arise on sections of the principal bundle of Appendix 2. We'll get to non-Abelian gauge connections like $U(1) \otimes SU(2)$ in future notebooks.

Gauge Covariant Derivative

Inputs

1. **expr**: The field being differentiated, scalar or tensor
2. **var**: the coordinate function, e.g., $x[u[\mu]]$, the differentiation variable

3. q : the coupling constant or “charge”
4. A : the gauge-field connection symbol. We keep Γ implicitly as the spacetime connection for all theories

```
In[9]:= CD[expr_, var_, q_, A_] :=
Module[{gravitationalPart, gaugeCorrection, idx},
  (*Calculate the coordinate Covariant Derivative via existing CD overloads.*)
  (*Handle  $\partial$  + Christoffels (if expr is a tensor)*)
  (*If expr is a scalar, just return  $\partial$ .**)
  gravitationalPart = CD[expr, var];
  (*Extract the index from the differentiation variable.*)
  (*var is some  $x[\mathcal{U}[\mu]]$ ; we need  $\mu$ *)
  idx = var[[1, 1]];
  (*Construct the Gauge Correction, e.g.,  $-i q A_\mu * \text{expr}$ *)
  (*Note: This A has a DOWN (Covariant) index matching the derivative*)
  gaugeCorrection = -I * q * A[D[idx]] * expr;
  (*Combine*)
  gravitationalPart + gaugeCorrection];
```

Electrodynamics Demos

Covariance of the Covariant Derivative

In “Forces as Gauge Connections,” we subjected the following expression to a 1D (Abelian) gauge transformation in a manual calculation. Now, include a specific, constant charge, q :

```
In[10]:= (cde = CD[ψ, x[ℳ[μ]], q, A])
Out[10]= -i q ψ (A_μ) + ψ_{,μ}
```

Let’s automate that manual calculation. Starting with the gauge transformations

$$\bar{\psi} = e^{iq\Lambda} \psi \quad \text{and} \quad \bar{A}_\mu = A_\mu + \partial_\mu \Lambda(x^\nu)$$

```
In[11]:= gaugeTransforms = {
  ψ → e^{iqΛ} ψ,
  A[D[μ_]] := A[D[μ]] + Partial[Λ, x[ℳ[μ]]]}
Out[11]= {ψ → e^{iqΛ} ψ, A_μ := A_μ + Λ_{,μ}}
```

prove that the covariant derivative

$$D_\mu \psi \equiv \partial_\mu \psi - iq A_\mu \psi$$

transforms exactly the same way as ψ

$$D_\mu \bar{\psi} = e^{iq\Lambda} D_\mu \psi$$

We just need one rule for the charge constant. Everything else is already in the toolkit’s

differentiationRules:

```
In[12]:= constantRules = {Partials[q, x[U[μ_]]] :-> 0}
```

```
Out[12]= {q, μ_ :-> 0}
```

```
In[13]:= transformedCde =
```

```
  cde /. gaugeTransforms //. differentiationRules /. constantRules // Expand //
  Simplify
```

```
Out[13]= e^{iqΛ} (-i q ψ (A_{μ}) + ψ_{,μ})
```

```
In[14]:= transformedCde === e^{iqΛ} cde
```

```
Out[14]= True
```

QED ■

Invariance of the Field-Strength Tensor

The field-strength tensor, $F_{\mu\nu}$, aka Faraday, is defined as follows

```
In[15]:= F[μ, ν] = Partials[A[D[ν]], x[U[μ]]] - Partials[A[D[μ]], x[U[ν]]]
```

```
Out[15]= -(A_{μ,ν}) + A_{ν,μ}
```

usually presented as $\partial_\mu A_\nu - \partial_\nu A_\mu$ to make the indices line up, but our notation checks out.

Now, subject it to the gauge transform:

```
In[16]:= F[μ, ν] /. gaugeTransforms
```

```
Out[16]= -(A_{μ} + Δ_{,μ})_{,ν} + (A_{ν} + Δ_{,ν})_{,μ}
```

See that it doesn't change by invoking the toolkit's `ExpandDerivatives`:

```
In[17]:= F[μ, ν] /. gaugeTransforms // ExpandDerivatives
```

```
Out[17]= -(A_{μ,ν}) + A_{ν,μ}
```

QED ■

Note: Charge Conservation as a Consequence

By proving that the covariant derivative transforms just by a phase factor, we proved that physical laws built from these derivatives remain invariant (unchanged) under gauge transformations.

In the Lagrangian, phase factors in terms like $\bar{\psi} D_\mu \psi$ cancel ($e^{iq\Lambda} * e^{-iq\Lambda}$), showing our theory has a continuous-in- Λ $U(1)$ symmetry (invariance). It's Abelian because the multiplication of phase factors is commutative.

According to Noether's Theorem, every continuous symmetry implies a conservation law.

- time-translation symmetry $\Rightarrow$ conservation of energy
- spatial translation symmetry $\Rightarrow$ conservation of momentum
- $U(1)$ phase symmetry $\Rightarrow$ conservation of electric charge

Thus, the “gauge correction” symmetry is the direct mathematical reason that electric charge is conserved.

Charged Vector Boson Covariance Demo

In this demo, we differentiate a charged contravariant vector field W^α , coupled to a neutral field Z_μ by a scalar hypercharge g_W . Z_μ acts like a vector phase correction in the non-geometric part of the covariant derivative. In addition to this non-geometric phase correction, W^α generates a geometrical correction proportional to Γ , as with any tensor.

We did not need a geometrical correction in electrodynamics because we were differentiating a scalar field ψ . Its geometrical covariant derivative is exactly equal to its ordinary partial derivative—the twisting and turning of spacetime does not change a scalar value. We do not need here a geometrical correction for Z_μ because it is not the subject of differentiation here.

This expression occurs in the electroweak, $U(1)$ sector, an Abelian sector (multiple phase operations commute), of the Standard Model of Particle Physics. In future notebooks, we will cover non-Abelian symmetries like those of $U(1) \otimes SU(2)$.

We proceed just as above

```
In[18]:= (wsd = CD[W[U[α]], x[U[μ]], gW, Z]) // TensorForm
Out[18]=

$$W^{\alpha}_{,\mu} - i g_W (W^\alpha) (Z_\mu) + (W^\lambda) \Gamma^\alpha_{\mu\lambda}$$


In[19]:= mixedGaugeTransforms = {
  W[U[α_]] -> e^{i g_W \Lambda} * W[U[α]],
  Z[D[μ_]] -> Z[D[μ]] + Partials[Λ, x[U[μ]]]}
Out[19]=

$$\{W^{\alpha_-} \rightarrow e^{i g_W \Lambda} (W^\alpha), Z_{\mu_-} \rightarrow Z_\mu + \Lambda_{,\mu}\}$$


In[20]:= transformedWsd =
  wsd /. mixedGaugeTransforms //. differentiationRules /. {Partials[gW, _] -> 0} //
  Expand // TensorForm
Out[20]=

$$e^{i g_W \Lambda} (W^{\alpha}_{,\mu}) - i e^{i g_W \Lambda} g_W (W^\alpha) (Z_\mu) + e^{i g_W \Lambda} (W^\lambda) \Gamma^\alpha_{\mu\lambda}$$


In[21]:= expectedWsd = wsd * e^{i g_W \Lambda} // Expand // TensorForm
Out[21]=

$$e^{i g_W \Lambda} (W^{\alpha}_{,\mu}) - i e^{i g_W \Lambda} g_W (W^\alpha) (Z_\mu) + e^{i g_W \Lambda} (W^\lambda) \Gamma^\alpha_{\mu\lambda}$$

```

```
In[22]:= transformedWsd === expectedWsd
Out[22]=
True

QED ■
```

Geometrical GR Demos

Covariant Derivative of a Scalar Function Λ

equal to its ordinary partial derivative, rendered in comma notation

```
In[23]:= CD[Λ, x[U[μ]]]
Out[23]=
Λ,μ
```

Covariant Derivative of a Contravariant Vector V^μ

```
In[24]:= CD[V[U[μ]], x[U[v]]] // TensorForm
Out[24]=
V^μ_{,v} + (V^λ) Γ^μ_{vλ}
```

Covariant Derivative of a (2, 0) Tensor

```
In[25]:= CD[T[U[μ], U[v]], x[U[ρ]]] // TensorForm
Out[25]=
T^μν_{,ρ} + (T^λν) Γ^μ_{ρλ} + (T^μλ) Γ^ν_{ρλ}
```

Covariant Derivative of a (1, 1) Tensor

```
In[26]:= CD[T[U[μ], D[v]], x[U[ρ]]] // TensorForm
Out[26]=
T^μν_{,ρ} - (T^μλ) Γ^λ_{νρ} + (T^λν) Γ^μ_{ρλ}
```

Covariant Derivative of the Covariant Metric

```
In[27]:= (cdg = CD[g[D[μ], D[v]], x[U[ρ]]]) // TensorForm
Out[27]=
g_{μν,ρ} - (g_{λν}) Γ^λ_{μρ} - (g_{μλ}) Γ^λ_{νρ}
```

Metric Compatibility, $\nabla g = 0$

$\nabla g = 0$ was an axiom in earlier notebooks, and it yielded the Christoffel symbols in the Fundamental Theorem of Riemannian Geometry. Here, we do the opposite, showing that the Christoffel symbols

imply metric compatibility. Thus, metric compatibility is both necessary and sufficient for the Christoffel symbols.

- In abstract geometric contexts, as opposed to coordinate-basis contexts, the Christoffel symbols are called the **Levi-Civita Connection**., often written $\left\{ \begin{smallmatrix} \lambda \\ \mu \nu \end{smallmatrix} \right\}$.
- Christoffel symbols were derived in Covariant Derivatives and Connections in the $\mathcal{UD}$ Calculus, assuming both torsion freedom (symmetry in lower two indices of Γ), and metric compatibility
- If we allow torsion, as in Einstein-Cartan theory or as in Appendix 1, the present tautology is not valid. In such cases, we must introduce the **contortion tensor** K

$$\Gamma^\lambda_{\mu\nu} = \left\{ \begin{smallmatrix} \lambda \\ \mu \nu \end{smallmatrix} \right\} + K^\lambda_{\mu\nu} \quad (8)$$

We do not develop this case further in this notebook.

For pedagogy, clear the definitions loaded from the toolkit:

```
In[28]:= ClearAll[ruleLeviCivita, cdgWithLeviCivitaConnection, metricDifferentiationRules];
```

Start with the covariant derivative of the metric, cdg , from above

Fetch the Levi-Civita connection via a new rule, added to Relativity Toolkit v1.9.0:

$$\Gamma^\lambda_{\mu\rho} = \frac{1}{2} g^{\lambda\sigma} (g_{\sigma\rho,\mu} + g_{\sigma\mu,\rho} - g_{\mu\rho,\sigma}) \quad (9)$$

```
In[29]:= ruleLeviCivita =  $\Gamma[\mathcal{U}[\lambda\_], \mathcal{D}[a\_], \mathcal{D}[b\_]] \Rightarrow$ 
Module[{ $\sigma = \text{Unique}["\sigma"]$ },
   $\frac{1}{2} g[\mathcal{U}[\lambda], \mathcal{U}[\sigma]] *$ 
  (Partials[g[ $\mathcal{D}[\sigma]$ ,  $\mathcal{D}[b]$ ], x[ $\mathcal{U}[a]$ ]] +
  Partials[g[ $\mathcal{D}[\sigma]$ ,  $\mathcal{D}[a]$ ], x[ $\mathcal{U}[b]$ ]] -
  Partials[g[ $\mathcal{D}[a]$ ,  $\mathcal{D}[b]$ ], x[ $\mathcal{U}[\sigma]$ ]])]
```

```
Out[29]=
```

$$\Gamma_{a_b_}^{\lambda_} \Rightarrow \text{Module}\left[\{\sigma = \text{Unique}[\sigma]\}, \frac{1}{2} (g^{\lambda \sigma 23}) (g_{\sigma 23 \ b,a} + g_{\sigma 23 \ a,b} - g_{a \ b, \sigma 23})\right]$$

In cdg , use this rule to replace Γ with its definition, then **Expand** to expose terms to further processing

```
In[30]:= (cdgWithLeviCivitaConnection = cdg /. ruleLeviCivita // Expand) // TensorForm
```

```
Out[30]=
```

$$-\frac{1}{2} (g^{\lambda \nu}) (g^{\lambda \kappa}) (g_{\kappa \mu, \rho}) - \frac{1}{2} (g_{\mu \lambda}) (g^{\lambda \kappa}) (g_{\kappa \nu, \rho}) -$$

$$\frac{1}{2} (g^{\lambda \nu}) (g^{\lambda \kappa}) (g_{\kappa \rho, \mu}) - \frac{1}{2} (g_{\mu \lambda}) (g^{\lambda \kappa}) (g_{\kappa \rho, \nu}) +$$

$$g_{\mu \nu, \rho} + \frac{1}{2} (g^{\lambda \nu}) (g^{\lambda \kappa}) (g_{\mu \rho, \kappa}) + \frac{1}{2} (g_{\mu \lambda}) (g^{\lambda \kappa}) (g_{\nu \rho, \kappa})$$

Now consider the following new Metric Differentiation Rules in the toolkit:

They include

- metric symmetry: $g_{\mu\nu} = g_{\nu\mu}$, require by standard general relativity
- metric contraction to the Kronecker delta: $g_{\lambda\nu} g^{\lambda\mu} = \delta^\mu_\nu$
- contraction of Kronecker delta on derivatives, for the chain rule: $\delta^\mu_\nu \partial_\mu = \partial_\nu$
- delta contraction inside metric derivatives: $\delta^\mu_\nu g_{\mu\lambda,\rho} = g_{\nu\lambda,\rho}$, $\delta^\mu_\nu g_{\lambda\mu,\rho} = g_{\lambda\nu,\rho}$

```
In[31]:= metricDifferentiationRules = {
  (*Metric Symmetry*)
  g[D[a_], D[b_]] /; ! OrderedQ[{a, b}] => g[D[b], D[a]],
  g[U[a_], U[b_]] /; ! OrderedQ[{a, b}] => g[U[b], U[a]],
  (*Permissive Contraction (All 4 Alignments)*)
  (*First-First*)
  g[D[s_], D[a_]] * g[U[s_], U[b_]] => delta[U[b], D[a]],
  (*Second-Second*)
  g[D[a_], D[s_]] * g[U[b_], U[s_]] => delta[U[b], D[a]],
  (*First-Second (Redundant but Safe)*)
  g[D[s_], D[a_]] * g[U[b_], U[s_]] => delta[U[b], D[a]],
  (*Second-First*)
  g[D[a_], D[s_]] * g[U[s_], U[b_]] => delta[U[b], D[a]],
  (*Delta Contractions*)
  delta[U[bound_], D[free_]] * Partial[expr_, x[U[bound_]]] =>
    Partial[expr, x[U[free]]],
  delta[U[bound_], D[free_]] * Partial[g[D[bound_], D[other_]], v_] =>
    Partial[g[D[free], D[other]], v],
  delta[U[bound_], D[free_]] * Partial[g[D[other_], D[bound_]], v_] =>
    Partial[g[D[other], D[free]], v]};
```

The result is immediate upon application of both the old metric rules and the new metric rules:

```
In[32]:= 0 ==
  (cdgWithLeviCivitaConnection //. (metricRules~Join~metricDifferentiationRules))
```

Out[32]=

True

QED ■

Riemann Revisited

Here is a concise derivation of the Fundamental Theorem of Riemannian Geometry:

Calculate the commutator: $[\nabla_\rho, \nabla_\nu] A^\mu$:

```

In[33]:= term1 = CD[CD[A[U[μ]], x[U[ν]]], x[U[ρ]]];
term2 = CD[CD[A[U[μ]], x[U[ρ]]], x[U[ν]]];
(comm = term1 - term2) // TensorForm

Out[35]=

$$\left( \left( A^{\lambda} \right) \Gamma_{\nu\lambda}^{\mu} \right)_{,\rho} - \left( \left( A^{\lambda} \right) \Gamma_{\rho\lambda}^{\mu} \right)_{,\nu} - \left( A^{\mu}{}_{,\lambda} \right) \Gamma_{\nu\rho}^{\lambda} + \left( A^{\mu}{}_{,\lambda} \right) \Gamma_{\rho\nu}^{\lambda} - \left( A^{\lambda} \right) \Gamma_{\nu\rho}^{\kappa} \Gamma_{\kappa\lambda}^{\mu} +$$


$$\left( A^{\lambda} \right) \Gamma_{\rho\nu}^{\kappa} \Gamma_{\kappa\lambda}^{\mu} - \left( A^{\lambda} \right) \Gamma_{\rho\lambda}^{\kappa} \Gamma_{\nu\kappa}^{\mu} - \left( A^{\lambda}{}_{,\rho} \right) \Gamma_{\nu\lambda}^{\mu} + \left( A^{\lambda} \right) \Gamma_{\nu\lambda}^{\kappa} \Gamma_{\rho\kappa}^{\mu} + \left( A^{\lambda}{}_{,\nu} \right) \Gamma_{\rho\lambda}^{\mu}$$


Simplification Pipeline

In[36]:= calculatedRiemann = (
    (comm //
      ExpandDerivatives //
      Expand //
      CanonicalizeIndices) //.
    torsionRules) //
    ExtractCoefficient[#, A] & //
    TensorForm

Out[36]=

$$\Gamma_{\lambda\nu,\rho}^{\mu} - \Gamma_{\lambda\rho,\nu}^{\mu} - \Gamma_{\lambda\rho}^{\kappa} \Gamma_{\kappa\nu}^{\mu} + \Gamma_{\lambda\nu}^{\kappa} \Gamma_{\kappa\rho}^{\mu}$$


Textbook definition:  $\Gamma_{\nu\bar{s},\rho}^{\mu} - \Gamma_{\rho\bar{s},\nu}^{\mu} + \Gamma_{\rho\lambda}^{\mu} \Gamma_{\nu\bar{s}}^{\lambda} - \Gamma_{\nu\lambda}^{\mu} \Gamma_{\rho\bar{s}}^{\lambda}$ 

In[37]:= rawTextbook = Partials[Γ[U[μ], D[ν], D[s]], x[U[ρ]]] -
    Partials[Γ[U[μ], D[ρ], D[s]], x[U[ν]]] +
    Γ[U[μ], D[ρ], D[λ]] * Γ[U[λ], D[ν], D[s]] -
    Γ[U[μ], D[ν], D[λ]] * Γ[U[λ], D[ρ], D[s]]

Out[37]=

$$\Gamma_{\nu\bar{s},\rho}^{\mu} - \Gamma_{\rho\bar{s},\nu}^{\mu} - \Gamma_{\rho\bar{s}}^{\lambda} \Gamma_{\nu\lambda}^{\mu} + \Gamma_{\nu\bar{s}}^{\lambda} \Gamma_{\rho\lambda}^{\mu}$$


In[38]:= expectedRiemann = (rawTextbook //. torsionRules) // TensorForm

Out[38]=

$$\Gamma_{\lambda\nu,\rho}^{\mu} - \Gamma_{\lambda\rho,\nu}^{\mu} - \Gamma_{\lambda\rho}^{\kappa} \Gamma_{\kappa\nu}^{\mu} + \Gamma_{\lambda\nu}^{\kappa} \Gamma_{\kappa\rho}^{\mu}$$


In[39]:= calculatedRiemann === expectedRiemann

Out[39]=
True

QED ■

```

Appendix 1: Spin Connection as Gravitational Gauge

Gravitation acts *almost* like any other gauge theory, but not quite. Gravitational forces arise from geometrical connections *in* the geometry, while forces in other gauge theories arise from non-gravitational connections *on* the geometry, connections in “wallpaper” spaces like $U(1)$ and $SU(3)$. This partial fit between gravitation and other gauge theories only deepens the frustration of unification-seekers.

To make gravitation a full gauge theory, we must find conserved quantities arising from local gauge

invariance. The definitive review paper is the following:

- Friedrich W. Hehl et al., *General Relativity with Spin and Torsion: Foundations and Prospects* (Rev. Mod. Phys. 48, 393, 1976)

which shows that:

- Invariance under general transformations (diffeomorphisms), implies conservation of the energy-momentum tensor.
- Invariance under local Lorentz transformations implies conservation of the spin angular momentum tensor (the spin current).
- Standard Einstein GR with vanishing torsion, which we assumed in this series of notebooks, collapses the two symmetries and hides the independent spin current.

Some other references are

- Sean Carroll, *Spacetime and Geometry: An Introduction to General Relativity*

- Appendix J, Page 486ff, under local Lorentz invariance, tetrads, and the spin connection:

Spin connection ω_μ^{ab} , arising from non-coordinate basis sets, acts as the gauge field for local Lorentz transformations. Carroll also explains that in standard GR the torsion tensor $T^\lambda_{\mu\nu}$ is zero, locking the Levi-Civita connection (Christoffel symbols) to the metric and forcing the energy-momentum tensor to be symmetric, absorbing spin effects via symmetrization.

- Also see Hehl, Equation 3.8 and Sections III.B and III.C. In other references, this may be called *Belinfante-Rosenfeld symmetrization*.

- T. W. B. Kibble, *Lorentz Invariance and the Gravitational Field* (J. Math. Phys. 2, 212, 1961)

- Foundational paper deriving gravitation as the gauge theory of the Poincaré group, i.e., Lorentz + translations

Kibble shows that (1) gauging translations in Poincaré yields the tetrad field, defining the metric and conserving energy-momentum, and that (2) gauging the Lorentz boost+rotation subgroup yields the spin connection, permitting transformation of spinors, otherwise impossible.

- Video: Carmeci Academy. *General Relativity | Review and Intro to The Spin Connection*, <https://www.youtube.com/watch?v=LjydxYNY7Yg>

Appendix 2: Fibre Bundles

In this appendix, we introduce the standard mathematical language. We present *bundles*, *projections*, *fibres*, *sections*, *connections*, and (formally) *invariance*, with each concept applied to both geometry/-gravitation and to more general gauge fields, with electrodynamics the motivating example.

TL;DR: Dictionary

Concept	Gauge Theory (electrodynamics)	Geometry / Gravitation (general relativity)	Interpretation and Analogy
Principal Bundle	Phase Bundle (P)	Frame Bundle (LM)	All Possible Wallpapers

Principal Section	Local Phase Choice	Local Frame Choice (Tetrad)	Smooth Wallpaper Choices
Structure Group	$U(1)$	$GL(4, \mathbb{R})$ or $SO(3, 1)$	Design Rules
Associated Bundle	Complex Line Bundle	Tangent Bundle (TM)	All Possible Walls
Associated Section	Charged Scalar Field	Vector Field V^μ	Particular Wall Choices
Connection	Vector Potential A_μ	Christoffel / Spin Connection	Wallpaper Glue

Bundles

Spaces attached to each point of a manifold are collected and flattened into sets called **bundles**. They are *fibre bundles* of projection functions (detail below).

In applications, there are two distinguished bundles: the **principal bundle** and the **associated bundle**, in a matched pair. Looking ahead, connections on sections of the principal bundle are “sources” of forces, and sections on associated bundles supply the quantities subject to forces, “victims” of forces.

Geometry / Gravitation

Let $T_p M$ (an indivisible notation) be the set of all tangent vectors (wit up indices) at point p of the manifold.

An **associated bundle** (victims of forces) of gravitation is the **tangent bundle** TM , the **disjoint union** of all points p in the manifold paired with all vectors v ripped out of their tangent spaces at p , flattened into one big set of pairs:

$$TM \equiv \bigsqcup_{p \in M} \{v \in T_p M\} = \{(p, v) \mid p \in M, v \in T_p M\} \quad (\text{associated bundle, victims of forces}) \quad (10)$$

- Likewise, the cotangent bundle $T^* M$ is the set of all points paired with all covectors (down indices) at each point, and so on for each valence (combination of up-down indices) of tensor. The other geometrical bundles, for tensors of other valences, can also be associated bundles (victims of forces).

The **principal bundle** (sources of forces) of gravitation is the **frame bundle**, which supplies, at each point, all possible bases for frames of reference.

- All possible bases include the ordinary coordinate bases that conceal spin and the tetrad bases that reveal spin, source of spin-coupled forces (Appendix 1).

Again, for emphasis, the **associated bundles** (victims of forces) are the associated tangent, cotangent, etc. bundles, which supply all possible tensors at each point.

Gauge Fields

With electrodynamics, imagine a copy of the symmetry group $U(1)$ (aka **structure group**) attached to each point $p \in M$. The members of $U(1)$ are phase factors, $e^{i\varphi}$ (abbreviated as φ), each representing a choice of gauge at that point. The **principal bundle** (sources of forces) of electrodynamics is the disjoint union of all manifold points p paired with all possible phase factors at p , flattened into one big set of pairs.

$$U(1)M \equiv \bigsqcup_{p \in M} \{e^{i\varphi} \in U(1)\} = \{(p, \varphi) \mid p \in M, e^{i\varphi} \in U(1)\} \quad (\text{principal bundle, sources of forces}) \quad (11)$$

- We might call $U(1)M$ a phase bundle.

In classical (non-quantum) electrodynamics, the **associated bundle** (victims of forces) is the **complex line bundle**, a collection of pairs of a point and a copy of the entire complex plane at that point, accounting for all possible values of a complex-valued **charged matter field**. This field represents density and internal alignment (physical interpretation of phase) of a notional charged fluid.

- The copies of the complex plane in the associated bundle is the entire complex plane, not just the members of $U(1)$, which are confined to the unit circle in the complex plane.

In quantum electrodynamics, the associated bundle is a collection of pairs of points and all possible Dirac wave functions, each of which is a superposition of spinor states. In this notebook, we cover only the classical case. The quantum case comes later in this series of notebooks.

Projections

Envision a **projection function**, π , that yields p from any element of TM :

$$\pi : TM \rightarrow M, (p, v) \mapsto p \quad (12)$$

“forgetting” the vector v .

In electromagnetism, the projection function $\pi : (U(1)M) \rightarrow M, (p, \varphi) \mapsto p$ forgets the phase factor $e^{i\varphi}$ from any given pair in $U(1)M$.

Fibres

In elementary mathematics, the **fibre**, $\text{fib}(f, p) \equiv f^{-1}(p)$, of a function f at some point p is the preimage of f at p , that is, the subset of the domain of f that yield p .

- In this usage, f^{-1} is not an “inverse function” that yields points, but a function that yields subsets. A general f may not be “invertible” in the usual sense of a point-to-point bijective function, but we can always define a fibre: a set containing all points of the domain that yield p . Unfortunately, we have the same notation, f^{-1} , both for an inverse function when f is invertible and for a fibre generally.

The fibres of the bundles—principal or associated—of interest to us are those of the projection functions π .

In geometry/gravitation, π^{-1} can’t “remember” the forgotten vector, so it must return the set of all vectors attached to p (or, more generally, tensors):

$$\pi^{-1}(p) = \{v \in T_p M\} \equiv T_p M \quad (\text{victims of forces}) \quad (13)$$

which is, of course, the entire tangent space at p .

In electrodynamics, the π^{-1} can’t remember the forgotten phase factor, so it must return a complete copy of $U(1)$ at each point:

$$\pi^{-1}(p) = \{e^{i\varphi} \in U(1)\} \equiv U(1) \quad (\text{sources of forces}) \quad (14)$$

Sections

Geometry / Gravitation

A **section of the principal bundle** (sources of forces, the frame bundle) is a particular choice of reference frame at each point—a coordinate basis for the Levi-Civita connection or a non-coordinate, tetrad basis for the spin connection.

A **section of the associated bundle** (victims of forces, the tangent bundle) is a smooth choice of one vector, $V^\mu(p)$, at each point, amounting to a smoothly varying vector field subject to forces. Likewise for other tensors.

Gauge Fields

In electrodynamics, a **section of the principal bundle** (sources of forces, the phase bundle) is a smooth choice of phase, $\varphi(p)$, at each point, which is a field of complex numbers of unit magnitude. It is the gauge-theory analog of a reference frame, a locally varying gauge choice.

A **section of the associated bundle** (victims of forces, matter fields) is a smooth choice of complex-valued charged matter field at each point (classical), or a smooth choice of Dirac wave function at each point (quantum).

Connections

Geometry / Gravitation

The connection on the gravitational principal bundle (sources of forces) is the field of Christoffel Symbols $\Gamma^\lambda_{\mu\nu}$, or the field of spin connections. These connections describe how an associated section—the vector field V^μ —twists and turns as p moves. A Levi-Civita connection is the Γ coefficient of the correction $V \cdot \Gamma$ that keeps the covariant derivative ∇V tensorial.

Gauge Fields

The connection on the principal gauge bundle (sources of forces) is the ***gauge potential***, $A_\mu(p)$, the field of vector potentials at each point. This field of connections describes how the associated section (victims of forces)—the charged matter field or the Dirac wavefunction field—twists and turns as p moves.

Just as $\Gamma^\lambda_{\mu\nu}(p)$ corrects for the twisting and turning of tangent vectors at each point p , $A_\mu(p)$ corrects for the twisting of the *internal phase* of the charged matter field at each point p .

Just as gravitational forces arise from the Levi-Civita or spin connection, electrodynamical forces arise from the gauge potential. We now finalize our slogan:

Forces arise from connections on sections of principal bundles.

Invariance

If the Lagrangian density, $\mathcal{L}$, of gravitation is invariant with respect to smooth frame choice at each point, the physics is **locally Lorentz invariant**.

If the Lagrangian density, $\mathcal{L}$, of electrodynamical gauge theory is invariant with respect to smooth choice of external unitary phase from $U(1)$, the physics is **locally gauge invariant**.

Invariant here means numerically equal and of the same algebraic form, as explained in my earlier notebook, *From the steam age to quantum fields*.

Conclusion

In this work, we bridge the conceptual and computational gap between general relativity and gauge theory. By treating “gauge” not just as mathematics but as a concrete choice of measurement basis—whether for position in spacetime or phase in a complex field—we unify the origin of forces under a single slogan: **Forces arise from connections on sections of principal bundles**.

We show that “garbage terms” generated by differentiating a phase-shifted field are mathematically analogous to “garbage terms” generated by differentiating a vector in curvilinear coordinates. In both cases, a connection field—the vector potential A_μ or the Christoffel Symbol $\Gamma^\mu_{\nu\lambda}$ —cancels these terms and restores phase covariance and geometrical covariance, respectively.

Computationally, the Relativity Toolkit v1.9.0 is now “doubly covariant.” It automatically produces identities for hybrid objects like the Charged Vector Boson W^α , which “ride on” spacetime geometry while twisting in an internal gauge space. This confirms that our symbolic machinery is ready for more of the Standard Model.

While this notebook focuses on the Abelian (commutative) $U(1)$ sector, the geometric machinery sets the stage for generalization to non-Abelian gauge theories incorporating $SU(2)$ and $SU(3)$.

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